Haze Damage Account

A detailed measurement framework for Southeast Asian transboundary haze including Singapore

Research and decision report
23 October 2026

Executive summary

A useful haze damage account should show **what was harmed, how much of the harm was caused by the specified fires, who bears the burden, and which interventions can prevent it**. One money figure cannot answer all four questions. The recommended design is a linked physical account, a financial incidence account, and a social welfare account, supported by a common evidence and overlap register.

The physical account records source-attributable $PM_{2.5}$ exposure, statistical health outcomes, disrupted work and education, protective resources, and other measured consequences. A separate source-fire extension records burned assets, ecosystems and greenhouse gases. The financial account traces expenditures, earnings changes, insurance, compensation and fines by payer and recipient. The welfare account values only a mutually compatible selection of impacts; its total is neither a bill nor a measured fall in GDP.

Measurement starts with a counterfactual: conditions with the specified fire emissions compared with the same weather and other emissions without those fires. Prevention appraisal needs a second comparison: realistic conditions with and without the proposed intervention. Removing all fires is not the benefit achievable by one project.

The strongest first implementation is a transparent pilot for one historical episode, with regional exposure and health estimates and a deeper Singapore account for health services, work, transport and household protection. Publish physical outcomes before monetization; report unmeasured categories explicitly. Expand the source-country and ecosystem accounts as access and evidence improve.

The worked example in this report is entirely hypothetical. It produces 9.72 statistical attributable deaths and a selected monetized social loss of **CU21.53 million**, using invented currency units and valuation inputs. An equally hypothetical intervention yields CU4.57 million of avoided damage for CU1 million in additional resources. These calculations teach the accounting method; they estimate no actual episode or country.

The central innovation opportunity is to connect prevention decisions to **expected avoided harm per additional dollar**, while retaining uncertainty, distribution and evidence of effectiveness. Hotspot counts, hectares treated, enforcement announcements and money collected remain useful operational measures, but require a demonstrated chain to reduced exposure and damage.

Reader guide and report structure

This report is a proposed analytical framework, informed by published haze studies and official health and economic assessment methods. It is not a systematic review, a completed regional damage estimate, or a legal determination. References identify which findings are empirical, modeled or methodological.

Pages	Content	Decision supported
3 to 4	Scope and evidence	What belongs in the account and what existing findings establish
5 to 7	Exposure and health	How specified fires become measurable population harm
8 to 11	Other impacts accounting and distribution	How sectors and affected groups enter compatible accounts
12 to 13	Data and uncertainty	What evidence is needed and how confidence is reported
14 to 16	Worked example and intervention appraisal	How quantities become values and project comparisons
17 to 18	Delivery plan and research priorities	How to build a useful first account and improve it
19 to 20	Technical appendices	Definitions and publication controls
21 onward	References	Primary research and official assessment methods

How to read the monetary results

Damage means an adverse change relative to a declared counterfactual. **Financial burden** means a change in expenditure, income, receipts or financing. **Social welfare loss** includes resource opportunity costs and defensible valuations of health risk and nonmarket harm. These lenses overlap; their headline totals must not be added.

Throughout the report, CU means invented currency units. A real account must declare the currency, constant price year, valuation date, exchange or purchasing-power conversion method, and discounting convention. Do not infer an empirical unit value from an illustration.

Recommended publication package

Publish a short decision brief, a physical outcome table, the financial incidence table, the selected welfare estimate with uncertainty, a source-fire extension, and an explicit list of omissions. Retain a reproducible technical workbook or database with source identifiers, formulas, scenario versions and overlap decisions. The report explains the specification for those future data products; it does not claim that they already exist.

1 Purpose scope and first principles

The question is not merely how much haze costs. A more useful question is: **which preventable changes in fire emissions cause the largest reduction in harm, for whom, at what cost, and with what confidence?** Start from mechanisms and observable quantities, then choose valuation methods that fit the decision.

Boundary	Recommended definition
Core smoke account	Health effects, visibility and activity disruption, protective resources and response caused by transported smoke. Include residual harm under actual behavior and protection.
Source-fire extension	Fire suppression and direct damage to assets, crops, peat, habitat and greenhouse gas emissions. Keep separate from smoke damage before presenting any combined total.
Geography	Show Indonesia, Malaysia, Singapore and other affected countries separately where data permit. Sum a declared regional boundary; report global climate effects as an extension.
Time	Distinguish the emission or episode period, the period when consequences occur, and the valuation date. Future effects require an appropriate causal model and stated horizon.
Population and sectors	Include residents and relevant transient populations consistently. Retain age, occupation, income and vulnerability strata when evidence and privacy allow.

Six questions that discipline the account

Clarify the claim. Define the event, emission sources, affected populations and the decision the estimate will inform. **Challenge assumptions.** Test whether observed illness, closures or lost revenue would have occurred without haze. **Require evidence.** Link each effect to a measured quantity or an explicitly labeled model.

Consider alternatives. Compare ambient versus personal exposure methods, acute versus life-table health models, and resource versus welfare valuation as alternatives where they overlap. **Examine consequences.** Check who gains or loses from intervention choices and whether risk shifts elsewhere. **Question the original metric.** A project that lowers hotspot counts may still leave substantial smoke exposure, especially where fires persist or emissions reach dense populations.

Define completion as an auditable, useful account of covered damage, rather than a claim that every possible harm has been monetized. Missing biodiversity, learning or wellbeing estimates must remain visible as omissions; they are not zero.

2 What the findings establish

Existing studies support an emissions-to-damage framework, but their results use different boundaries, years and assumptions. They should inform the account rather than supply interchangeable coefficients.

Evidence	Finding and implication for this account
World Bank 2015 [1]	A broad Indonesia assessment estimated fire and haze damage and losses of about US$16.1 billion. Its sector and health coverage differs from later modeled estimates. Preserve the original boundaries.
Kiely and colleagues 2021 [2]	Combines fire emissions, regional health and land-related costs with restoration scenarios. Modeled restoration of 2.49 million hectares reduced burned area by 6% and PM emissions by 24% in the studied 2015 scenario. This is a modeled scenario, not observed project effectiveness.
Marlier and colleagues 2019 [3]	Links land cover, fire emissions, atmospheric transport and population health to restoration targeting. It supports ranking locations by downwind health consequences, not treating each hectare as equally valuable.
Falcon and colleagues 2022 [4]	A village randomized trial of a particular incentive package found no statistically significant fire reduction, despite improved prevention effort. Spending or participation alone cannot establish avoided damage.
ProbFire and hydrological modeling [5,6]	Retrospective work shows useful forecasting skill in some settings and potential gains from hydrology. Forecast accuracy still needs an operational test of whether action reduces harm.
Recent restoration evaluations [7,8]	Observational studies estimate context-dependent effects of restoration or canal blocking. Drought, hydrological coverage and comparison design matter. Transfer effects with explicit uncertainty.
Peat fire field evidence [9]	GAMBUT experiments document persistent smoldering despite rainfall. A disappearance of surface flame is insufficient evidence of extinguishment or zero emissions.

The World Bank and Kiely totals substantially overlap and must not be added. Differences can arise from scope, health horizons and valuation choices rather than contradictory observations. Kiely used a carbon-price approach in its historical calculation; that is not the same as measuring marginal climate damage. [1,2]

Research on fire-related land transactions also informs the incentive analysis in Section 9. Its findings describe actor relationships and historical transaction allocations; they are distinct from estimates of damage to people and ecosystems. [26]

3 Attribute pollution and measure exposure

Construct the relevant counterfactual

For a historical episode, compare actual surface concentrations with modeled concentrations after removing specified fire emissions, holding meteorology and other emissions constant. Specify whether the sources are all regional landscape fires, peat fires, a geographic subset or another defined group. A chemical transport model or validated source-apportionment method supplies the fire contribution; monitors anchor the receptor concentration estimates. [3,10]

$$\Delta C_{it} = C_{it}^{\text{actual}} - C_{it}^{\text{cf}}$$

Equation 1 Source-attributable concentration increment

Here i is a location, t a time period and cf the counterfactual. Use concentration in μg/m³ with an averaging period matched to the health model. For nonlinear atmospheric chemistry, removal of separate sources can yield effects that do not sum to the joint removal effect. If an additive allocation is required, declare the allocation rule rather than silently rescaling results.

Publish interpretable exposure measures

$$E = \sum_{i,\,t} P_{it}\,\Delta C_{it}\,\Delta t$$

$$\overline{\Delta C}_{PW} = \frac{\sum_{i,\,t} P_{it}\Delta C_{it}\Delta t}{\sum_{i,\,t} P_{it}\Delta t}$$

Equation 2 Total exposure and population and time weighted mean increment

P is the relevant population and Δt is duration in days. E has units person·μg/m³·day; the weighted mean remains μg/m³. Neither is a count of harmed people or a measure of absorbed dose. Report concentration distributions, source contributions and person-days above clearly defined thresholds alongside the mean. Define a unique-person convention before reporting people exposed.

Use $PM_{2.5}$ concentrations rather than PSI as a substitute exposure variable. An air quality index is a communication indicator, not the concentration input for a $PM_{2.5}$ risk function. Satellite aerosol optical depth, hotspot counts and burned area each measure different parts of the chain. A hotspot is neither a measured emission mass nor a ground-level exposure. [11,12]

For Singapore, combine NEA monitoring or historical readings with modeled regional fire contributions. Record station representativeness, calibration, missing hours, daily aggregation and model validation. Do not apply an arbitrary mask or indoor-infiltration reduction to an ambient concentration-response function without a consistent supporting model. [11,13]

4 Estimate attributable health outcomes

Choose each health endpoint and its evidence before calculating costs: for example, all-cause mortality for a supported age range, asthma admissions, respiratory consultations or symptom-days. Functions must match the pollutant, outcome, population, concentration range, exposure averaging period and lag structure. Use smoke-specific evidence where robust and transferable; document any use of all-source $PM_{2.5}$ evidence. [14,15]

Keep the baseline rate convention explicit

Let r^{actual} be the incidence rate under actual exposure, expressed per person for the modeled period; RR(C) is the applicable relative risk function. The attributable outcome expectation is:

$$\Delta N = \sum_{i,a,t} P_{iat} r_{iat}^{\text{actual}} \left[1 - \frac{RR_a(C_{it}^{\text{cf}})}{RR_a(C_{it}^{\text{actual}})}\right]$$

For a log-linear function and a published relative risk RR_{10} per 10 µg/m³, this becomes:

$$\beta = \frac{\ln(RR_{10})}{10}, \qquad \Delta N = \sum_{i,a,t} P_{iat} r_{iat}^{\text{actual}} [1 - e^{-\beta \Delta C_{it}}]$$

If the available rate is instead a counterfactual no-haze rate, use the corresponding form below. The two rate conventions cannot be exchanged. [15]

$$\Delta N = \sum_{i,a,t} P_{iat} r_{iat}^{\text{cf}} \left[\frac{RR_a(C_{it}^{\text{actual}})}{RR_a(C_{it}^{\text{cf}})} - 1\right]$$

Age stratum a must match the response function and incidence data. Annual rates require appropriate time conversion for a daily calculation; seasonal or outbreak differences may make a simple division inadequate. For nonlinear risk functions, evaluate risk at both total concentrations rather than applying the function only to the smoke increment.

Choose one consistent mortality framework

Short-term time-series effects and long-term cohort or life-table effects answer different questions. A short intense episode cannot automatically be treated as the full effect of sustained lifetime exposure. For recurring haze, specify the exposure trajectory, cessation lag and follow-up horizon. Report alternative acute and long-term estimates separately when their endpoints overlap. [14,16]

Attributable outcomes are statistical expectations for a population. They do not identify individual deaths or diagnoses certified as caused by haze. Do not add all-cause deaths to their cause-specific components, or several alternative risk-function estimates. Independent pollutant burdens also need an overlap assessment.

5 Measure and value health loss

Keep health quantities visible

$$\mathrm{DALY} = \mathrm{YLL} + \mathrm{YLD}$$

$$\mathrm{YLL} = \sum_a \Delta\mathrm{Deaths}_a L_a^*$$

$$\mathrm{YLD}_{\mathrm{episode}} = \sum_j \Delta\mathrm{Cases}_j \, d_j \, DW_j$$

A disability-adjusted life year combines years of life lost and years lived with disability. L* is standard remaining life expectancy at the relevant age; d is duration in years and DW a disability weight. The episode-based YLD equation is an incident-episode approach. Standard annual burden accounts use prevalent health states and comorbidity adjustment; do not mix the two conventions without reconciliation. Standard WHO GHE DALYs do not apply age weighting or time discounting. [17]

Estimating YLL from acute deaths needs care about mortality displacement and the remaining life expectancy of affected people. Assigning every acute attributable death the population average remaining lifetime can overstate certainty. Use an appropriate life-table approach or present the limitation explicitly. [16,17]

Value mortality risk with a declared method

The value of a statistical life, VSL, aggregates willingness to pay for small mortality-risk reductions. It is not a payment for an identified life. Multiply statistical attributable mortality by an appropriate VSL for a selected welfare estimate. A life-year or monetized DALY approach is an alternative valuation path; do not add it to VSL for the same outcomes. Do not add lost lifetime earnings to a comprehensive mortality-risk welfare valuation. [18,19]

For cross-country results, publish the chosen transfer method, income elasticity, currency basis and price year. Present a common regional VSL sensitivity alongside any income-adjusted national approach so the influence of valuation on geographic priorities is visible. Retain physical health outcomes as the common comparison. This is a recommended equity sensitivity, not an assertion that one monetary convention settles ethical choices.

Separate illness resources from nonpecuniary harm

Measure additional consultations, admissions, medicines, treatment time and caregiving with incremental unit resource costs. Audit whether a cost-of-illness value already includes lost work. Likewise, a willingness-to-pay value for illness may already include treatment, earnings, discomfort and leisure losses. Add separate rows only when their components are demonstrably nonoverlapping. [20]

Protective behavior reduces some exposure while consuming resources. Combine damage remaining under actual behavior with its incremental protective resource cost. Pairing an unprotected health scenario with the costs of protection that prevented part of that harm would overstate the actual burden.

6 Quantify work household and sector effects

Estimate changes against a credible no-haze baseline, net of substitution, postponed activity and catch-up. Difference-in-differences, event studies or other causal designs may support attribution when their assumptions and diagnostics are defensible. Raw differences between smoky and clear periods are not sufficient. [21,22]

Category and quantity	Valuation and principal control
Work and business Excess absent hours; output per hour; interrupted production	Use attributable net value added or an appropriate surplus measure. Do not add worker wages, firm revenue and sector output for the same lost activity. Count mortality earnings only in a separate compatible income view.
Transport and tourism Delay hours; cancellations; visitor nights; extra operations	Measure additional real operating resources, defensible time costs and net activity loss. Travel shifted across dates or countries may redistribute spending rather than destroy the full regional amount.
Education Student-days disrupted; instructional hours; learning changes; caregiver time	Record closures physically. Value actual catch-up resources and nonoverlapping caregiver opportunity cost. Model persistent learning and earnings effects only with causal evidence; closure days are not themselves lifetime income loss.
Household protection Masks filters electricity cleaning and time	Count the increment caused by haze. Allocate durable equipment to services consumed during the event, accounting for residual use and value. Do not count both full purchase cost and the same future service costs.
Public response Staff time monitoring communications shelters and logistics	Use incremental resources consumed rather than total agency budgets. Distinguish cash disbursement from staff opportunity cost and other resource costs.
Wellbeing and recreation Restricted activity; symptoms; sleep or validated wellbeing measures	Retain physical or survey measures. Monetize only with suitable evidence and a defined overlap boundary; broad wellbeing values can overlap health discomfort, leisure and recreation losses.

Observed spending is not automatically social harm. Added clinical revenue can accompany the loss of resources used to treat avoidable illness. A cancelled purchase may be deferred or replaced elsewhere. The financial incidence table records those changes; the welfare calculation values the net consequences within its chosen boundary. [19,21,22]

For a first Singapore module, prioritize data that can be linked to dates and exposure: aggregated health utilization, employer absence, school disruption, flight operations and household protection spending. Mark inaccessible data as missing and document any proxy rather than imputing a precise total without support.

7 Add the source fire and ecosystem extension

Direct burning and smoke are connected, but they cause distinct damage. Publish the following extension beside the core smoke account. Combine totals only after reconciling scope, time and valuation. Stock damage, production losses and additional costs require separate treatment. [21,22]

Impact	Physical account and valuation approach
Built assets and crops	Record destroyed or damaged assets, crop area and expected output. State the depreciated asset, repair or replacement basis. Reconstruction funding needs need not equal welfare asset loss. Avoid adding capitalized income value and the same future income stream.
Peat and land	Record burned area, burn depth, peat volume or mass and land condition with uncertainty. Map drainage and hydrological units as well as parcels. Distinguish direct fire damage from prior degradation.
Ecosystems and biodiversity	Measure habitat extent, condition, fragmentation and relevant species or population changes. Value selected final ecosystem services where supported. Preserve unpriced biodiversity indicators and recovery trajectories.
Suppression and restoration	Record real additional firefighting, rehabilitation and monitoring resources. Separate preventive investment from event damage; distinguish expenditure from demonstrated ecological recovery.
Climate	Estimate incremental emissions by greenhouse gas, accounting for relevant baseline emissions and subsequent recovery. Report tonnes of each gas, emission year and uncertainty before valuation.

$$D_{\text{climate}} = \sum_{g} E_g \, SC_g$$

Equation 3 Gas specific emissions multiplied by compatible social cost values

E_g is incremental tonnes of gas g and SC_g its selected marginal social cost per tonne, matched to emission year, currency, price year, geographic scope and discount convention. A carbon tax, allowance or credit price is a policy or financing price, not automatically a damage value. Applying a CO_2 social cost to generic CO_2-equivalent emissions is only a labeled approximation. The cited 2023 EPA report supplies a methodological example, not a statement of current policy. [24]

UN SEEA Ecosystem Accounting supplies extent, condition, service and asset accounts. Its monetary exchange values are not comprehensive welfare values; keep them separate from willingness-to-pay estimates unless reconciled. Do not add intermediate ecosystem services already embodied in crops or timber, or an ecosystem asset value and the services capitalized into it. Avoid valuing the same carbon release again as lost carbon stock. [23]

8 Reconcile the three linked accounts

Account	Records	Meaning of its total
Physical outcomes	Source-attributable exposure; illness and mortality; time and output; assets ecosystems and emissions	Multiple units remain visible. Do not sum deaths, days and hectares into an arbitrary index without explicit weights.
Financial incidence	Expenditure receipts income changes insurance compensation fines and grants by payer and recipient	Shows cash and financing burdens. Some entries are transfers; others finance the same resources recorded elsewhere.
Social welfare	Selected nonoverlapping resource opportunity costs and nonmarket valuations	Shows a valuation of lost wellbeing. Its resource subset is already included and is not an extra additive headline.

Maintain an overlap register

Assign each effect an endpoint identifier, an overlap group and a flag showing whether it enters the selected total. Document why alternative estimates are excluded. The same treatment episode may appear in hospital activity, insurer payments and household copayments, but contributes one resource estimate with a financing split.

Potential overlap	Required decision
Acute and chronic mortality; all-cause and cause-specific deaths	Choose a consistent primary framework. Show overlapping alternatives separately.
VSL monetized YLL or DALY and mortality earnings	Choose a compatible valuation path. Never add them for the same deaths.
Morbidity willingness to pay and treatment or work loss	Read the unit value definition and include only components it excludes.
Absence wages revenue and sector value added	Select the net production or welfare loss and reconcile distribution separately.
Local tourism loss and regional gain; assets and capitalized flows	Net substitution within the social boundary and remove repeated income streams.

Fines, compensation, grants and insurance mainly redistribute purchasing power within a defined social boundary. They neither erase the physical damage nor create a second damage item. Record real legal, administrative and remediation resources separately. A transfer across the boundary can matter to a national financial account while canceling in the regional account. [25]

An estimated welfare loss expressed as a percentage of GDP is a scale comparison, not evidence that GDP fell by that percentage. Additional health spending can even raise measured activity while reflecting harm. Macroeconomic model results require reconciliation with sector estimates before addition. [19,22]

9 Distribution incentives and accountability

For every major endpoint, identify affected people, organizations using resources, funders, recipients and the actors able to change the source of risk. Publish country and subgroup results before aggregation. Include outdoor workers, children, older adults, people with relevant pre-existing disease, lower-income households and communities near fires when the data support the distinction. Do not infer vulnerability from nationality alone.

Measure incentives and enforcement outcomes

An accountability module should link investigations to verified decisions, final sanctions, appeal status, collection, compliance and subsequent fire outcomes. Cases, suspects, companies and parcels are different units; record-level linkage is needed before calculating progression or resolution rates. Treat this register as a proposed measurement tool, with documented definitions and missing records, rather than assuming that an investigation count demonstrates effective deterrence.

Purnomo and colleagues document a historical local land-trading arrangement, including US$856 per hectare for ready-to-plant land and US$486 allocated to an organizer. These context-specific transaction allocations help identify actor incentives; they are not haze damage, current national averages or estimates of net profit. Use the original study to distinguish those receiving transaction benefits from those bearing health and environmental costs. [26]

Evaluate an enforcement or incentive change against a credible comparison, using subsequent compliance, fire recurrence, emissions and damage as outcomes. Record administrative and prevention resources separately from fine receipts and compensation. Mapped fires and management information can guide investigation, but do not by themselves establish a particular actor’s misconduct.

Measurement layer	Record and interpret
Scientific attribution	Estimated emission source and contribution to exposure or harm, with uncertainty and model version.
Control and incentives	Verified land rights management duties contracts financing and prevention costs. Separate documented facts from hypotheses.
Legal outcomes	Verified determinations and procedural status; distinguish allegation investigation judgment appeal and collection.
Behavioral response	Changes in ignition drainage compliance or recurrence after a policy, evaluated against a credible comparison.

An atmospheric model does not establish who ignited a fire, controlled a parcel, breached a duty or is legally liable. Shared drainage and transported smoke also cross boundaries. Keep legal conclusions separate from source-contribution estimates. The policy question is whether incentives reduce expected regional damage, not merely whether a penalty exceeds one historical worker payment.

10 Build an auditable data system

Use a versioned set of linked tables rather than one opaque total. The minimum analytical grain is a geography, period, population stratum, endpoint and scenario. Preserve raw data and transformations so every published number can be traced back to an input and a method.

Table	Required contents and likely access
Episode and scenarios	Event dates source scope geography no-target-fire scenario intervention assumptions weather version and time horizon. Public specification.
Concentrations and sources	Grid-day or station-hour $PM_{2.5}$; fire inventory transport model source tags quality flags and validation. NEA historical data and modeled regional contributions are complementary. [11,13]
Fire and land	Detected hotspots burned area fuel peat depth drainage hydrology land cover and uncertainty. NASA FIRMS detects thermal anomalies; note detection limits and false or missed detections. [12]
Population and health	Spatial and age-stratified population; endpoint-specific incidence rates; response coefficients averaging periods lags and provenance. Clinical data often require controlled access.
Sector and household effects	Activity measures counterfactual model incremental resources unit costs and substitution assumptions. Access agreements may be needed for employers transport and household microdata.
Valuation and overlap	Unit values currency price year valuation basis transfer method scope included components overlap group and selected-total flag.
Incidence and uncertainty	Payer recipient transfers subgroup allocation probability distributions correlations model alternatives and evidence status.

Minimum fields in each effect record

Store event and scenario IDs; source scope; geography; period; subgroup; endpoint; quantity and unit; lower, central and upper values where justified; causal method; baseline ID; valuation method; unit value; currency and price year; valuation date and discount rate; payer and recipient; accounting lens; overlap group; inclusion flag; exclusion reason; source; and model version. Appendix A defines key variables.

Privacy and missingness

Use aggregated or appropriately protected health and employment data, with data-owner agreements and documented disclosure controls. Public outputs should not identify patients or employees. Track whether a blank cell means unavailable, not applicable, not estimated, below a reporting threshold or true zero. Those states have different meanings and must survive aggregation.

11 Quantify uncertainty and validate attribution

Four different sources of uncertainty

Parameter uncertainty includes emission factors, fuel burned, baseline disease rates, response coefficients and unit values. **Structural uncertainty** includes transport models, causal specifications and health-response forms. **Policy scenarios** cover drought, adoption, maintenance and displacement. **Missing harm** is an omitted category; it cannot be represented simply by a narrow interval around the measured subtotal.

Use probabilistic propagation where defensible, retaining material correlations. A single risk-coefficient draw should apply consistently across relevant populations; treating each grid cell as an independent epidemiological study creates false precision. Preserve dependencies between fire severity, emissions, exposure and sector disruption. Record which uncertainty sources are included in a reported interval.

Checks before publication

Check	Evidence of an acceptable result
Exposure	Validation against appropriate monitors; out-of-sample performance where possible; residual patterns by location and smoke intensity; documented gaps.
Health	Correct endpoint age band time unit baseline-rate convention and lag; no out-of-range extrapolation hidden; comparison of plausible response functions.
Sector causality	Control for season temperature humidity holidays trends respiratory outbreaks and other relevant confounders. Examine pre-trends and placebo tests where applicable.
Accounting	No repeated endpoints transfers or capitalized flows; currencies price years horizons and units reconciled; subgroup totals aggregate correctly.
External validity	State how geography drought land condition behavior and institutions differ from the evidence source. Distinguish modeled effectiveness from field evaluation.

A low-central-high valuation table is a sensitivity exercise unless it is explicitly generated as a probabilistic interval. Do not call it a 95% confidence interval. Publish exposure-model alternatives, health-function alternatives and valuation sensitivities separately when collapsing them into one range would obscure their meaning.

Retain fractional expected cases during calculation and round only displayed results. Report enough precision for reproducibility without suggesting that a modeled death count or monetized total is known exactly. A quantified subtotal can be useful even when the complete burden is unknown; make its coverage prominent.

12 Worked example from exposure to health

All numbers on pages 14 to 16 are invented for teaching. The example represents no actual Singapore or regional haze event. The risk coefficient is not an epidemiological recommendation. Assume a constant population, concentration and daily baseline, a same-day response, no lag and no seasonal variation.

Input	Hypothetical value
Eligible population	1,000,000 people
Episode duration	30 days
Actual daily mean $PM_{2.5}$	30 μg/m³ every day
Counterfactual daily mean $PM_{2.5}$	10 μg/m³ every day
Actual-scenario annualized mortality rate	600 deaths per 100,000 people per year
Invented short-term RR_{10}	1.01 per 10 μg/m³

The excess concentration is 20 μg/m³. Total source-attributable exposure is 1,000,000 × 20 × 30 = **600 million person·μg/m³·days**. The population-and-time-weighted mean increment is 20 μg/m³. These exposure measures do not by themselves indicate a number of illnesses or deaths.

$$r_{\text{day}}^{\text{actual}} = \frac{600}{100{,}000 \times 365} = 0.00001643836$$

Under these simplifying assumptions, actual-scenario expected mortality over 30 days is 1,000,000 × 30 × 0.00001643836 = 493.1507 deaths. Convert the invented RR to a coefficient and calculate the attributable fraction:

$$\beta = \ln(1.01)/10 = 0.0009950331$$

$$AF = 1 - e^{-\beta \times 20} = 1 - 1/1.0201 = 0.01970395$$

$$\Delta N = 493.1506849 \times 0.01970395 = 9.7170167$$

The model expectation is **9.72 statistical attributable premature deaths**, with counterfactual expected mortality of 483.43. This does not identify which people died because of haze. Use the unrounded expectation in the monetary calculation. In a real study, justify the health function and lag rather than adopting these teaching assumptions.

13 Worked example of the monetary account

Continue the invented example using the unrounded mortality expectation. Assume exactly 2,000 attributable nonfatal episodes, explicitly separate surviving-worker production effects, and the hypothetical resource and valuation inputs below. CU denotes constant-price invented currency units.

Selected component	Assumption	CU million
Mortality-risk loss	9.7170167 statistical deaths × CU2 million VSL	19.43
Nonfatal treatment resources	2,000 episodes × CU200	0.40
Nonfatal pain and discomfort	2,000 episodes × CU300; nonpecuniary only	0.60
Lost production	Separate surviving-worker events; net of substitution and catch-up	0.80
Protective resources consumed	Event-attributable services and materials; residual value allowed for	0.20
Transport resources and nonwork delay	No overlap with production loss illness time or other rows	0.10
Selected social loss	**Sum of compatible included components**	**21.53**

The resource and opportunity-cost subset is CU1.50 million: treatment 0.40 + production 0.80 + protection 0.20 + transport/time 0.10. Mortality-risk and nonfatal nonpecuniary valuation contribute CU20.03 million. The resource subset is already inside CU21.53 million. No lost mortality earnings or monetized DALYs are added. Education, biodiversity and other omitted categories remain unquantified.

The illness unit value here is defined to exclude medical bills, earnings and separately counted time. That assumption makes the addition valid in this illustration. A real willingness-to-pay estimate may not have this property and must be audited. [20]

Financial incidence is a different view

Suppose a CU0.30 million fine is paid by a company to a government within the regional boundary. Record the company payment and government receipt, linked to the same transfer ID. Physical outcomes and the selected CU21.53 million social loss are unchanged. Actual legal administration uses resources; actual remediation can change future damage. Both need their own records. [25]

For treatment, split the CU0.40 million financing across households, insurers and government only when data exist. Those shares reconcile to the financing total; they are not additional treatment resources. Lost household earnings and firm sales may describe incidence but cannot simply be added to the selected production estimate.

14 Value prevention and compare interventions

For project j, estimate expected damage without it and damage with it under realistic effectiveness, adoption, drought, maintenance and displacement. Benefits are the difference. Include implementation, monitoring, enforcement, maintenance, opportunity costs and relevant adverse effects in the project resource cost, avoiding duplication with the damage calculation.

$$B_{jt} = D_t^{\text{without}} - D_t^{\text{with } j}$$

$$NPV_j = \sum_t \frac{B_{jt} - K_{jt}}{(1+r)^t}, \qquad BCR_j = \frac{\sum_t B_{jt}/(1+r)^t}{\sum_t K_{jt}/(1+r)^t}$$

K is additional intervention resource cost and r is the stated social discount rate. Financial returns require a separate revenue and payment model. A high social benefit-cost ratio does not mean that an investor can collect those benefits as cash.

Sensitivity and intervention example

Invented valuation scenario	Lower	Central	Higher
VSL in CU million per statistical death	1.00	2.00	4.00
Baseline selected social loss in CU million	11.82	21.53	40.97
Avoided damage in CU million	2.57	4.57	8.57
Additional intervention resource cost in CU million	1.00	1.00	1.00
Benefit-cost ratio	2.57	4.57	8.57
Net social benefit in CU million	1.57	3.57	7.57

Hold physical outcomes and effectiveness fixed across these valuation scenarios. Assume the intervention avoids 2 statistical deaths, 500 illness episodes, CU0.20 million production loss, CU0.10 million protection cost and CU0.02 million transport/time loss. Avoided illness treatment and discomfort are CU0.10 million and CU0.15 million. Central benefits total 4.00 + 0.10 + 0.15 + 0.20 + 0.10 + 0.02 = **CU4.57 million**.

Central residual damage is CU16.96 million; adding CU1 million project cost gives CU17.96 million, compared with CU21.53 million without the project. The net gain is CU3.57 million. No discounting is needed in this single-period illustration. The table is not a confidence interval or evidence of real effectiveness; a real appraisal must vary effectiveness, duration and costs as well as valuation.

15 Build a first account in ninety days

A ninety-day program can deliver a scoped pilot account if the required data and expertise are available. It cannot establish new chronic disease, learning or ecosystem causal relationships within that period. Use existing validated methods, document gaps and plan subsequent research.

Period	Deliverable and acceptance condition	Lead expertise
Days 1 to 15	Agree the event source and social boundaries; separate attribution and intervention counterfactuals; define endpoints and overlap register; secure data-owner commitments.	Program lead atmospheric science economics epidemiology
Days 16 to 35	Acquire and harmonize monitoring source emissions population health sector and expenditure data. Establish versioned provenance missingness and privacy rules.	Data engineering and data owners
Days 36 to 55	Validate source-attributable exposure; estimate supported health outcomes; publish an internal physical account and main uncertainty drivers.	Atmospheric science and epidemiology
Days 56 to 70	Estimate priority sector and household effects; build financial incidence; apply selected values with an explicit reconciliation.	Economics sector analysts and statisticians
Days 71 to 80	Propagate parameter uncertainty; compare model and valuation alternatives; test regional substitution overlap leakage and distribution.	Methods leads and independent reviewer
Days 81 to 90	Resolve review findings; publish coverage table physical outcomes selected monetary results source-fire extension and reproducible methods.	Program lead independent review and communications

Minimum useful release

Deliver a defensible $PM_{2.5}$ exposure account, one consistent mortality framework, supported nonfatal health endpoints, priority incremental resources and a selected monetary subtotal. State country and subgroup coverage, causal assumptions, omitted damage and uncertainty. A Singapore-focused sector module can coexist with a regional health module if their different coverage is prominent.

Decision ownership

Assign named owners for scope changes, model releases, epidemiological transfer, unit values and overlap decisions. Require an independent atmospheric and health methods check and an economic reconciliation. Retain a change log so revisions in a headline total can be explained by new data, changed boundaries or changed values.

16 Highest impact research and innovation priorities

The account becomes most useful when it changes a prevention decision. The priorities below are proposed opportunities inferred from the evidence, rather than proven products or guaranteed effects. Evaluate them with the same outcome and counterfactual discipline as other interventions.

Target sources by expected harm avoided

Connect hydrological units, land condition, emissions, weather and downwind populations to expected marginal damage. Compare rewetting, ignition prevention, rapid suppression and exposure protection using feasible avoided emissions and health effects. Marlier and Kiely show why area treated alone can miss large differences in downwind benefit. Field and quasi-experimental evidence must anchor realized effectiveness. [2,3,7,8]

Close the forecast to action loop

Test whether early warnings change staffing, water management, suppression speed or exposure protection, and whether those actions reduce emissions or health burden. Forecast skill is an intermediate outcome. Record false alarms, missed events, response cost, lead time and prevented harm. Retrospective forecasting results justify trials, not claims of demonstrated damage reduction. [5,6]

Test incentives against the whole causal chain

Trace who receives land-clearing or trading benefits, who can alter drainage or ignition, and who bears regional damage. Compare a defined incentive or enforcement change with an appropriate control. Measure prevention behavior, emissions and recurrence, not only disbursement or participation. The village trial’s null fire result shows why a plausible incentive mechanism still needs outcome evaluation. [4,26]

Make accountability and distribution auditable

Link scientific source estimates to verified management duties and separate legal records. Track enforcement progression, collection and behavioral response. Pair efficiency rankings with a distribution table so the needs of source-area communities and highly exposed workers remain visible even where incomes or measured market activity are lower.

Resolve the most consequential unknowns first

Prioritize evidence whose plausible range could change the investment decision: avoided peat combustion, intervention durability, source-to-receptor exposure, smoke-specific health response and distribution. Use value-of-information analysis to compare the expected gain from resolving an uncertainty with research cost. Do not devote equal effort to every minor cost category while a major effectiveness assumption remains unsupported.

The recommended success criterion is a verified reduction in expected and realized harm for the resources used, with transparent effects on affected groups. A complete-looking dollar total is less useful than a reproducible account that makes its coverage and decision consequences clear.

Appendix A Variable dictionary and calculation rules

Variable	Definition and unit	Control
C actual and C cf	Actual and counterfactual surface pollutant concentrations in μg/m³	Same averaging period weather and non-target emissions; source scope declared
ΔC	Difference between actual and counterfactual concentration in μg/m³	Retain model uncertainty; nonlinear chemistry can complicate additive source allocation
P and Δt	Eligible population in people and interval duration in days	Age geography and time must match incidence and response functions
E	Population-integrated concentration increment in person·μg/m³·day	An exposure indicator not absorbed dose or health outcome
r actual or r cf	Endpoint incidence per person per calculation period	Choose the correct rate convention and time conversion
RR and β	Endpoint risk function; log-linear coefficient per μg/m³	Record population pollutant lag range and source study
ΔN	Attributable expected outcomes in cases or statistical deaths	Retain fractional values internally; avoid overlapping endpoint totals
L star d and DW	Standard remaining life expectancy; episode duration in years; disability weight	Do not mix incident and prevalent burden conventions
VSL	Value per statistical mortality-risk reduction in declared currency	Record price year income transfer and included components
E gas and SC gas	Incremental tonnes by gas and compatible social cost per tonne	Match gas emission year currency and discount convention
D B K and r	Damage avoided benefits intervention resource costs and social discount rate	Same scenario boundary price basis and time horizon
CU	Invented currency units used only in the worked example	No conversion to actual country estimates is implied

Quantities and values should be independently reviewable. Store both the unrounded result and displayed precision. An alternative model estimate should use a distinct scenario or method identifier, not replace the original result without a version record.

Appendix B Reporting and audit specification

Required headline tables

Coverage. For each endpoint show geography, period, population, measured or modeled status, monetary inclusion, and reason for omission. **Physical outcomes.** Report source-attributable exposure, health and activity measures with units and appropriate ranges. **Financial incidence.** Show payer, recipient, expenditure or transfer type and linked effect. **Welfare.** List included components, alternative estimates, overlap exclusions and the selected subtotal. **Source-fire extension.** Keep direct assets, ecosystems and climate effects distinct before reconciliation.

Release checks

Question	Pass condition
Can a reviewer reproduce every headline?	Each total links to versioned inputs formulas unit values and selected-total flags.
Are the counterfactuals clear?	Historical source attribution is separate from realistic intervention effectiveness.
Do the accounts reconcile?	Financing shares do not create extra resources; transfers cancel within the stated boundary; no unsupported addition of ledger totals.
Are endpoints and units consistent?	Age bands incidence periods lags concentrations currency bases and time horizons match.
Are competing methods visible?	Acute and chronic health alternatives; model structures; valuation and equity sensitivities are clearly labeled.
Are unknowns represented honestly?	Missing harm is listed separately from zero and from quantified parameter intervals.
Is intervention value credible?	Adoption maintenance leakage durability costs and distribution are tested; social benefit is separate from cash return.
Are responsibility claims supported?	Scientific source estimates management information and legal findings remain distinct.

How to interpret a revision

A revised total may reflect stronger haze, a larger affected population, better health evidence, broader geographic coverage or changed valuation. Publish a bridge from the previous estimate that attributes the change to quantities, coverage, methods and unit values. Do not present a methodological revision as an observed worsening or improvement in actual haze.

This report supplies an accounting design and a transparent teaching example. Producing an actual regional estimate requires assembling the data, selecting context-appropriate risk functions and unit values, validating source attribution and completing the reconciliation described above.

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